

EE595H (Stochastic Models): Half Semester Examination

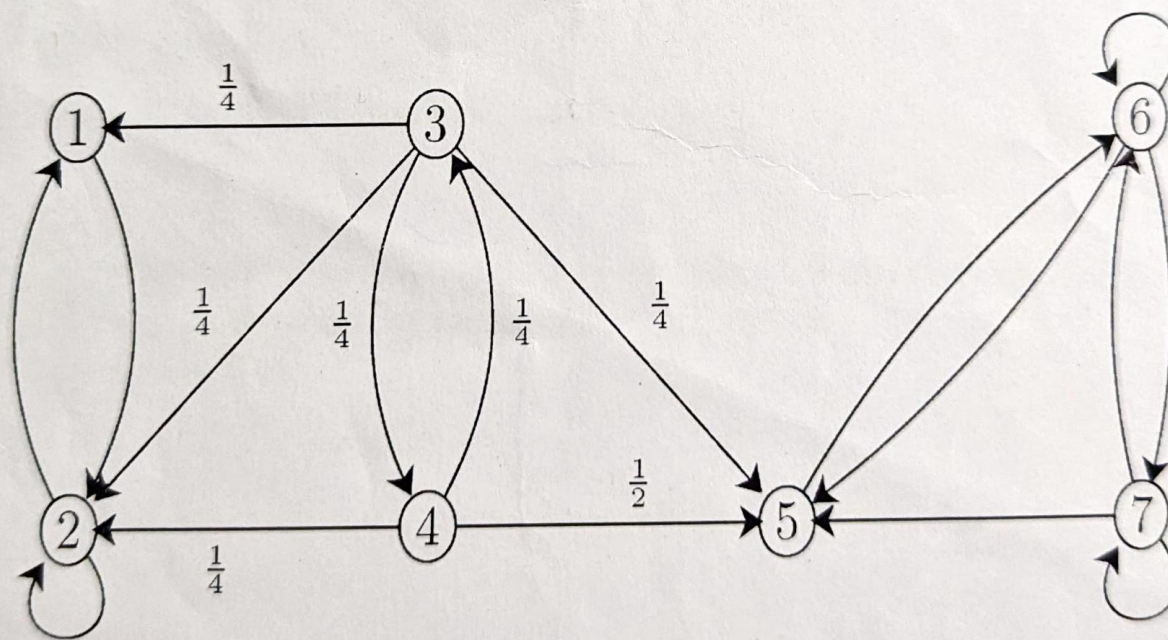
Total Questions: 4, Total Marks: 40, Time: 120 Mins

1. The following is the transition probability matrix of a Markov chain with state space $\{0, 1, 2, 3, 4\}$.

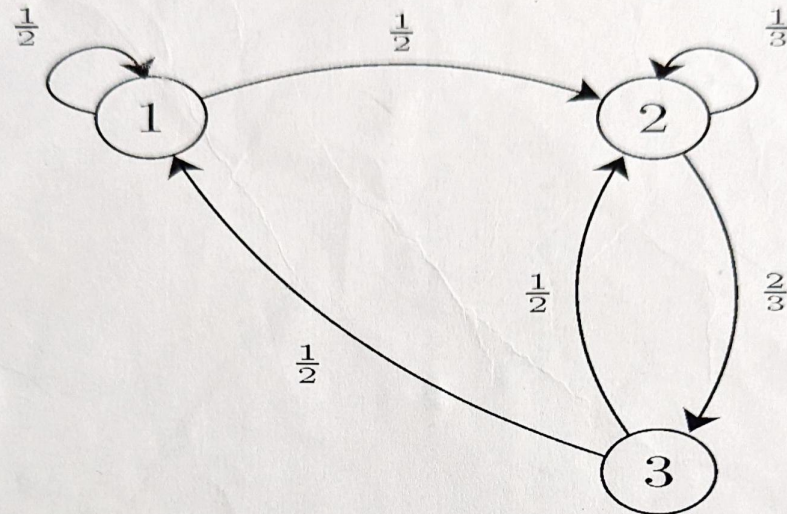
$$P = \begin{pmatrix} 2/5 & 0 & 0 & 3/5 & 0 \\ 1/3 & 1/3 & 0 & 1/3 & 0 \\ 0 & 0 & 1/2 & 0 & 1/2 \\ 1/4 & 0 & 0 & 3/4 & 0 \\ 0 & 0 & 1/3 & 0 & 2/3 \end{pmatrix}$$

Specify the classes, and determine which classes are transient and which are recurrent.

2. Consider the Markov chain in the following figure. There are two recurrent classes, $R_1 = \{1, 2\}$, and $R_2 = \{5, 6, 7\}$. Assuming $X_0 = 4$. Find the probability that the chain gets absorbed to R_1 .



3. Consider the Markov chain shown in the following figure.



- (a) Is this chain irreducible?
- (b) Is this chain aperiodic?
- (c) Find the stationary distribution for this chain.

2

2

6

Total: 10

4. Develop an algorithm to generate a random variable having the cumulative distribution function (CDF) by using the inverse transform method:

$$F_X(x) = \frac{x^2 + x}{2}; \quad 0 \leq x \leq 1$$

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